

Smart Parking App

Risk Management and Communication Plan | \$oftware ¢orp

1. Risk Identification

This section lists the risks that could threaten the Smart Parking App project. The risks are grouped into four categories: technical, schedule, financial, and people, with four or more each.

Technical Risks

- **TECH-01** Heavy traffic load causing slow updates or app crashes
- **TECH-02** Payment errors, such as failed or double charges
- **TECH-03** Sensor or IoT outages in parking lots
- **TECH-04** Incorrect data, such as wrong availability or pricing

Schedule Risks

- **SCHED-01** Delays caused by city approvals
- **SCHED-02** Late hardware delivery
- **SCHED-03** Longer-than-expected testing sessions
- **SCHED-04** Scope creep

Financial Risks

- **FIN-01** Payment processor fees eating into margins
- **FIN-02** Losing the city contract
- **FIN-03** Hidden costs from hardware maintenance
- **FIN-04** Going over budget

People Risks

- **PPL-01** Team or stakeholder issues
- **PPL-02** Key developer leaving the project
- **PPL-03** Poor communication between teams
- **PPL-04** Low driver adoption
- **PPL-05** Operators resisting the new system

2. Risk Register

The register below tracks each risk with its likelihood, impact, owner, and planned response. The project manager reviews it weekly and updates the status column as things change.

Probability and impact are each rated on three levels: low, medium, and high. Response strategies follow the four standard types, which are avoid, mitigate, transfer, and accept.

ID	Risk Description	Probability	Impact	Owner	Response Strategy	Status / Notes
TECH-01	Heavy traffic load causes slow updates or crashes	Medium	High	Lead Developer	Mitigate: load test before launch, add autoscaling, cache availability data.	Open. Peak demand expected near events and downtown rush hours.
TECH-02	Payment errors such as failed or double charges	Medium	High	Payments Engineer	Mitigate: use a certified processor, add reconciliation checks, build a clear refund flow.	Open. Ties to the dispute resolution gap found in research.
TECH-03	Sensor or IoT outages in parking lots	Medium	High	Infrastructure Lead	Mitigate: monitor sensor health, show last known data with a flag, keep spare units.	Open. Needs a fallback plan before pilot lots go live.
TECH-04	Incorrect data shows wrong availability or pricing	High	High	QA Lead	Mitigate: add data validation, cross-check sensors, show a last updated time.	Open. Targets the real-time availability gap in competitor research.
SCHED-01	City approvals take longer than planned	High	Medium	Project Manager	Mitigate: submit permits early, keep regular contact with city staff, add schedule buffer.	Open. Depends on city review calendar.
SCHED-02	Hardware arrives late	Medium	High	Procurement Lead	Mitigate: order early, line up a backup supplier, track shipments weekly.	Open.
SCHED-03	Testing sessions run longer than expected	High	Medium	QA Lead	Mitigate: reserve extra test time, automate regression tests.	Open.
SCHED-04	Scope creep from new feature requests	High	Medium	Project Manager	Avoid: use a change control process and lock requirements after SRS sign-off.	Open. Review any new request against the SRS.
FIN-01	Payment processor fees eat into margins	Medium	Medium	Finance Lead	Mitigate: negotiate rates, compare processors, build fees into pricing.	Open.
FIN-02	Losing the city contract	Low	High	Account Manager	Mitigate: hit contract milestones, send regular progress reports, keep the city relationship strong.	Monitoring.
FIN-03	Hidden hardware maintenance costs	Medium	Medium	Finance Lead	Mitigate: get maintenance quotes up front, keep a reserve, check warranty terms.	Open.

ID	Risk Description	Probability	Impact	Owner	Response Strategy	Status / Notes
FIN-04	Project goes over budget	Medium	High	Project Manager	Mitigate: hold monthly budget reviews and keep a contingency reserve.	Open.
PPL-01	Team or stakeholder issues	Medium	Medium	Project Manager	Mitigate: hold regular check-ins, set clear roles, escalate early.	Open.
PPL-02	Key developer leaves the project	Low	High	Engineering Manager	Mitigate: document code, cross-train developers, use pair programming.	Monitoring.
PPL-03	Poor communication between teams	Medium	High	Project Manager	Mitigate: follow the communication plan and use one shared task tracker.	Open. See Communication Plan section.
PPL-04	Low driver adoption	Medium	High	Marketing Lead	Mitigate: simple onboarding, launch promotions, clear upfront pricing.	Open.
PPL-05	Operators resist the new system	Medium	Medium	Operations Lead	Mitigate: involve operators early, offer training, start with pilot lots.	Open.

3. Communication Plan

This plan explains how each team meets and how project updates reach the right people. Clear communication also reduces several of the people and schedule risks listed in the register above.

Meeting Cadence by Team

Each team follows a fixed meeting schedule so problems surface early and decisions do not stall.

Team	Meeting	Cadence	Purpose
Development	Daily standup	Daily, 15 min	Share progress and surface blockers
Development	Sprint planning and review	Every 2 weeks	Plan the next sprint and demo finished work
QA and Testing	Test sync	Twice weekly	Review defects and test progress
Hardware and Infrastructure	Status check	Weekly	Track deliveries, installs, and sensor health
Operations and Support	Operator feedback sync	Weekly	Review operator issues and driver complaints
Finance	Budget review	Monthly	Compare spend to budget and review processor fees
Project Management	Risk review	Weekly, 30 min	Update the risk register and assign actions
All team leads	Cross-team leads meeting	Weekly	Align teams and resolve cross-team dependencies
Executives and city stakeholders	Steering committee	Monthly	Review progress, approvals, and contract status

Reporting Methods

Reports move up through set channels, and each report has a clear audience and frequency.

Report	Frequency	Audience	Method
Weekly status report	Weekly	Project team and management	Email and shared drive
Risk register update	Weekly	Project manager and team leads	Shared spreadsheet
Sprint review summary	Every 2 weeks	Developers, product, stakeholders	Live demo and meeting notes
Budget report	Monthly	Finance and executives	Dashboard and PDF summary
Stakeholder progress report	Monthly	City representatives	Email and slide presentation
Incident alert	As needed	Developers and operations	Team chat channel and on-call phone
Daily team communication	Ongoing	All team members	Team chat channels by team

Escalation

Issues a team cannot resolve on its own move to the project manager first, then the steering committee.